

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0174

Hinton Ap 345 kV
February 02, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0174. The Interconnection Customer proposes a 1195.2 MW natural gas combined cycle facility interconnecting at the Hinton Ap 345 kV substation in the West Virginia area. The request was received on November 15, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 131 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$74,947,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Aspen 345/138kV Transformer is overloaded to 173.9% of its 407 MVA emergency rating for the loss of the Crandall – Aspen 345 KV. The project contributes 14.5% DFAX (173.6 MW) to this overload.

The Hinton Ap Substation 345 kV/138 kV Auto is overloaded to 167.6% of its 346 MVA emergency rating for the loss of the Lowmoor – Hinton Ap 345 KV. The project contributes 38.3% DFAX (457.7 MW) to this overload.

The Dayton – Harrisonburg 138 KV is overloaded to 149.5% of its 377 MVA emergency rating for the loss of the Dayton Substation – Chatom 345 KV. The project contributes 8.5% DFAX (101.1 MW) to this overload.

Chatom – Purdy 138 KV: 147.6% loading (279 MVA) following the outage of the Dpc Prairie Du Chien – Frito Lay 138 KV; DFAX = 5.3%.

Villa Mar 2 1658 – Aspen 138 KV: 144.9% loading (227 MVA) following the outage of the Aspen Substation – Crandall 345 KV Line; DFAX = 7.3%.

Crandall – Carodean 138 KV line: 134.6% loading (236 MVA) following the outage of the Crandall – Dayton Substation 345 KV L; DFAX = 6.6%.

The Villa Mar 2 1658 – Orchard 138 KV is overloaded to 134.3% of its 234 MVA emergency rating for the loss of the Crandall – Aspen Substation 345 KV Line. The project contributes 7.3% DFAX (87.0 MW) to this overload.

For the loss of Dayton 345/138 kV Transformer, the Chatom 345 kV/138 kV Transformer loads to 124.0% of its 494 MVA rating. Project DFAX: 8.4%.

For the loss of Crandall – Aspen 345 KV, the Orchard Substation – Declo 138 KV loads to 123.8% of its 244 MVA rating. Project DFAX: 7.3%.

For the loss of Dayton – Chatom 345 KV, the Dayton 345/138 kV Autotransformer loads to 123.7% of its 479 MVA rating. Project DFAX: 8.5%.

For the loss of Video Pd – McAuley Substation 138 KV, the Declo – New London 138 KV loads to 116.5% of its 307 MVA rating. Project DFAX: 5.4%.

The Crandall 345/138 kV Transformer Bank is overloaded to 105.9% of its 375 MVA emergency rating for the loss of the Crandall – Dayton 345 KV. The project contributes 7.0% DFAX (84.2 MW) to this overload.

The Lowmoor 345-138 kV Transformer is overloaded to 105.6% of its 359 MVA emergency rating for the loss of the Aspen – Crandall 345 KV. The project contributes 10.6% DFAX (127.2 MW) to this overload.

Bear – Clover 345 KV: 102.3% loading (2,416 MVA) following the outage of the Mecklenburg Power Station – Person 345 KV line; DFAX = 9.2%.

Greenland Gap – Santee Cooper 345 KV: 102.2% loading (1,470 MVA) following the outage of the Spindle 500-345 kV Transformer; DFAX = 5.8%.

Snarr – Rhinelander Paper Substation 138 KV: 100.5% loading (203 MVA) following the outage of the Lowmoor Substation – Hinton Ap 345 KV; DFAX = 11.1%.

The Glen Lyn – Snarr 138 KV is overloaded to 100.0% of its 213 MVA emergency rating for the loss of the Lowmoor – Hinton Ap 345 KV. The project contributes 11.1% DFAX (132.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aspen 345/138kV Transformer	Replace transformer with higher-rated unit	\$16,132,000	26% (\$4,253,000)
Hinton Ap 345-138 kV TX	Replace transformer with higher-rated unit	\$14,692,000	87% (\$12,723,000)
Dayton – Harrisonburg 138 KV	Rebuild line with higher-capacity conductor	\$2,036,000	42% (\$866,000)
Chatom – Purdy 138 KV	Rebuild line with higher-capacity conductor	\$5,954,000	46% (\$2,724,000)
Villa Mar 21658 – Aspen 138 KV Line	Rebuild line with higher-capacity conductor	\$3,097,000	35% (\$1,080,000)
Crandall – Carodean 138 KV	Rebuild line with higher-capacity conductor	\$3,833,000	43% (\$1,644,000)
Villa Mar 2 1658 – Orchard 138 KV Line	Rebuild line with higher-capacity conductor	\$5,337,000	35% (\$1,861,000)
Chatom 345/138kV TX	Replace transformer with higher-rated unit	\$13,392,000	43% (\$5,773,000)
Declo Substation – Orchard 138 KV	Rebuild line with higher-capacity conductor	\$11,826,000	35% (\$4,124,000)
Dayton 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,912,000	42% (\$7,190,000)
Declo – New London 138 KV	Rebuild line with higher-capacity conductor	\$51,005,000	22% (\$11,117,000)
Crandall 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$12,089,000	40% (\$4,809,000)
Lowmoor 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$16,352,000	39% (\$6,399,000)
Bear – Clover 345 KV	Reconductor line	\$32,958,000	14% (\$4,745,000)
Greenland Gap – Santee Cooper 345 KV	Reconductor line	\$3,063,000	61% (\$1,881,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rhineland Paper – Snarr Substation 138 KV	Reconductor line	\$2,954,000	82% (\$2,435,000)
Gln Lyn – Snarr 138 KV	Reconductor line	\$1,605,000	82% (\$1,323,000)
	Total allocated to project		\$74,947,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.